AtiNAMatiC-Double -- User Manual

Version 0.1.0 · Atij 666 Studio · Windows (Standalone, VST3 & CLAP)

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1. What it is

AtiNAMatiC-Double is a **studio double-tracking simulator** designed to sit after your amp/cab simulator in the signal chain. It produces two virtual "performance" voices from a single take, panned to opposite sides, with independent slow random-walk pitch and timing drift -- giving the perceptual impression of two separate performances without any audible chorus artefacts or comb filtering.

Key design goals:

- **Never sounds like a chorus.** The modulation is a band-limited random walk updating roughly every 0.5--1.5 seconds per voice, not a sine LFO.
- **Mono-safe.** The dry signal is always blended back through the centre so a sum to mono cannot phase-cancel to silence.
- **Always outputs stereo** -- works from a mono or stereo input.
- **One macro.** The WIDTH knob controls delay spread, drift magnitude and pan amount simultaneously through a smooth S-curve so the "natural double" zone sits in the comfortable 30--60% range.

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2. Signal chain placement

```
Guitar  ->  Amp Sim (NAM / Helix / etc.)  ->  Cab / IR  ->  [Reverb/Delay]  ->  * AtiNAMatiC-Double *  ->  Master
```

Place Double **after** cab/IR and any mono reverb/delay. It should be one of the last inserts before the master bus so the stereo image it creates is the final spatial statement of the track.

For parallel double-tracking (blend the "doubled" image under a dry lead):

```
Lead guitar (dry) ------->|
                          +-> Mix -> Master
Doubled send (AtiNAMatiC-Double, MIX 100%) -->|
```

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3. Controls

WIDTH (large centre knob)

The master macro. Simultaneously scales:

- **Base delay spread** -- 8/11/14 ms (voice A) and 14/19/26 ms (voice B) depending on Mode, all scaled further by WIDTH.
- **Drift magnitude** -- how far each voice's pitch and timing wanders around its base delay.
- **L/R pan amount** -- voice A pans left, voice B pans right; at WIDTH 100% voice A is hard left, voice B is hard right.

A live readout (e.g. "45 %") is displayed at the knob centre.

Tick labels below the knob:

Label	Range	Character
MONO	0--15%	Almost no separation; use as a subtle thickener
DOUBLE	15--70%	Natural double-tracking sweet spot
WIDE	70--100%	Deliberately wide, more dramatic stereo image

MODE (Tight / Loose / Wide)

Shapes the base delay times and maximum drift swing. Does **not** change the rate of the random walk -- all modes stay slow.

Mode	Voice A base	Voice B base	Drift swing	Max pan
Tight	8 ms	14 ms	1.2 ms	85%
Loose	11 ms	19 ms	2.0 ms	90%
Wide	14 ms	26 ms	2.8 ms	100%

- **Tight** -- studio close-mic double, very coherent, suits rhythm guitars and bass.
- **Loose** -- classic "two takes, different day" spread; the default for most uses.
- **Wide** -- maximum spread, almost M/S-like; good for leads and ambient textures.

MIX (0--100%)

Wet/dry balance. At 100% you hear the full doubled image with the dry centre blend baked in. Lower it to taste to keep a leaner dry sound under the image.

OUTPUT (-24 to +12 dB)

Final trim after all processing. The doubling can add perceived loudness; use this to level-match against your bypass.

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4. Stereo Image panel

Located below the mode picker, the STEREO IMAGE panel gives real-time visual feedback:

Level bars (L / R) -- RMS dB bars from -60 dBFS to 0 dBFS with colour coding:

- Cyan: normal signal
- Amber: approaching 0 dB
- Red: at or above 0 dB

Voice dot field -- a horizontal strip showing voice A (cyan dot, left-biased) and voice B (violet dot, right-biased). Both dots drift around their pan centres as the random walk updates, giving a live picture of the current stereo spread.

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5. System requirements

Item	Minimum
OS	Windows 10 64-bit (or later)
CPU	Any modern x64 processor

RAM	64 MB per instance
Plugin formats	VST3, CLAP (DAW); Standalone (.exe)
Sample rates	44.1 / 48 / 88.2 / 96 / 176.4 / 192 kHz
Latency	Zero samples (no lookahead)
DAW	Any VST3- or CLAP-compatible host

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6. Installation

Windows -- Installer (recommended)

1. Run **AtiNAMatiC-Double-Setup-0.1.0.exe**.
2. Select the formats you want (Standalone, VST3, CLAP).
3. Click Install.
4. **Rescan plug-ins in your DAW** (see table below).

Default install paths:

Format	Location
Standalone	`C:\Program Files\Atij 666 Studio\AtiNAMatiC-Double\AtiNAMatiC-Double.exe`
VST3	`C:\Program Files\Common Files\VST3\AtiNAMatiC-Double.vst3`
CLAP	`C:\Program Files\Common Files\CLAP\AtiNAMatiC-Double.clap`

Windows -- Manual install

Copy the files from `Installer\Bundle\` to the paths above.

macOS -- Manual install

```
Standalone:  /Applications/AtiNAMatiC-Double.app
VST3:        ~/Library/Audio/Plug-Ins/VST3/AtiNAMatiC-Double.vst3
AU:          ~/Library/Audio/Plug-Ins/Components/AtiNAMatiC-Double.component
CLAP:        ~/Library/Audio/Plug-Ins/CLAP/AtiNAMatiC-Double.clap
```

Build from source: `cmake -S . -B build && cmake --build build --config Release`

Linux -- Manual install

```
VST3:  ~/.vst3/AtiNAMatiC-Double.vst3
CLAP:  ~/.clap/AtiNAMatiC-Double.clap
```

Build from source: `./build-linux.sh`

DAW rescan after install

DAW	How to rescan

Reaper	Options > Preferences > VST > Re-scan
Cubase / Nuendo	Studio > VST Plug-in Manager > Update
Studio One	Options > Locations > VST Plug-ins > Reset Blacklist + Rescan
Ableton Live	Preferences > Plug-Ins > Rescan
Bitwig Studio	Settings > Plug-ins > Rescan plug-ins
FL Studio	Options > Manage plugins > Find more plugins
Logic Pro (macOS)	Restart Logic; AU auto-validates on launch

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7. Controls reference

Control	Range	Default	Unit
WIDTH	0 -- 100	45	%
MODE	Tight/Loose/Wide	Tight	--
MIX	0 -- 100	100	%
OUTPUT	-24 -- +12	0	dB

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8. Internal signal chain

```
Input (mono or stereo)
  |
  v
[Mono sum (L+R)*0.5]
  |
  +-----> [Voice A delay line]  <-- slow random-walk drift (rate ~0.7-1.6 s)
  |            |  fractional-sample read tap
  |            v
  |       [Pan A: left-biased]
  |
  +-----> [Voice B delay line]  <-- independent random-walk drift (rate ~0.9-1.8 s)
  |            |  fractional-sample read tap
  |            v
  |       [Pan B: right-biased]
  |
  +-----> [Dry centre blend]  (always present, scales with WIDTH for mono safety)
  |
  v
[MIX crossfade with dry input]
  |
  v
[OUTPUT gain trim]
  |
```

```
  v
Stereo output
```

All delay taps use linear interpolation and a slewed read position (the slew itself creates the micro-pitch variation). There is no nonlinear processing and therefore no oversampling requirement.

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9. Typical use cases

Rhythm guitars (hard double)

- WIDTH: 70%, MODE: Tight, MIX: 100%
  Tight base delays with hard pan give a classic "guitars left and right" image from one track.

Natural double (two-take feel)

- WIDTH: 45%, MODE: Loose, MIX: 100%
  The sweet spot -- sounds like two separate performances.

Subtle thickener (mono mix)

- WIDTH: 15%, MODE: Tight, MIX: 100%
  Very little separation; mainly adds density without obviously stereo-ising the track.

Wide lead guitar

- WIDTH: 60%, MODE: Wide, MIX: 80%
  Spread lead over the stereo field; back off MIX slightly to keep a centred anchor.

Parallel double-tracking

- Send a copy of the track to AtiNAMatiC-Double, WIDTH: 80%, MIX: 100%.
  Blend the return under the dry lead at -6 to -12 dB.

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10. Tips

- **Start with MODE: Loose at WIDTH 40--50%** -- this is the most natural starting point for most tracks.
- **Watch the stereo image panel** -- if the voice dots drift too wide apart the image may collapse oddly on mono playback; bring WIDTH down until they sit within the centre 60% of the strip.
- **Automate WIDTH** for a build-up: start a verse at 30%, push to 60% in the chorus for a dramatic stereo expansion.
- **Do not use after reverb.** Applying double-tracking to a reverb tail causes audible smearing. Put reverb after Double, not before.
- **MIX at less than 100%** is useful when blending with a dry lead; this lets you keep a tight centre while still getting separation on the sides.
- **Output trim**: doubling two voices at 0.55 gain each adds roughly +1.5 dB perceived loudness at WIDTH 50%. Use OUTPUT -1.5 dB to level-match for A/B comparison.

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11. About

AtiNAMatiC-Double is part of the **AtiNAMatiC Series** -- a family of focused, hardware-flavoured guitar tools for the modern post-amp signal chain.

- Design & development: **ATIJ** (Atij 666 Studio)
- Built with **JUCE 8** (juce.com)

- CLAP support via **clap-juce-extensions** (free-audio)
- Built with MSVC / CMake
- Thanks to the open-source community

Personal and commercial use permitted on machines you own. Bug reports and feature requests: click the **AtiNAMatiC-Double** title badge in the plug-in window to open the About screen.

AtiNAMatiC-Double -- Instant double tracking. No chorus. No comb.